CLARKSON UNIVERSITY

PROJECT

FEVER & VITAL SIGNS EXPLORER

AN INTERACTIVE SHINY DASHBOARD

A PROJECT REPORT

SUBMITTED BY:

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Submitted in partial fulfilment of the requirements for the course

IA 640: Information Visualization

UNDER GUIDANCE OF

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MASTER OF SCIENCE IN APPLIED DATA SCIENCE
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A. Dataset Overview

The dataset is designed to assist in predicting recommended medications for patients based on their fever condition, symptoms, medical history, and other relevant factors. It incorporates a mix of patient health data, environmental variables, and lifestyle choices to improve model accuracy and better simulate real-world scenarios.

Dataset Characteristics:
Total Samples: 1000 (modifiable based on user needs).
Number of Features: 19 features + 1 target column.
File Format: CSV (enhanced_fever_medicine_recommendation.csv).

A1) Explanation of variables ?

Answer:

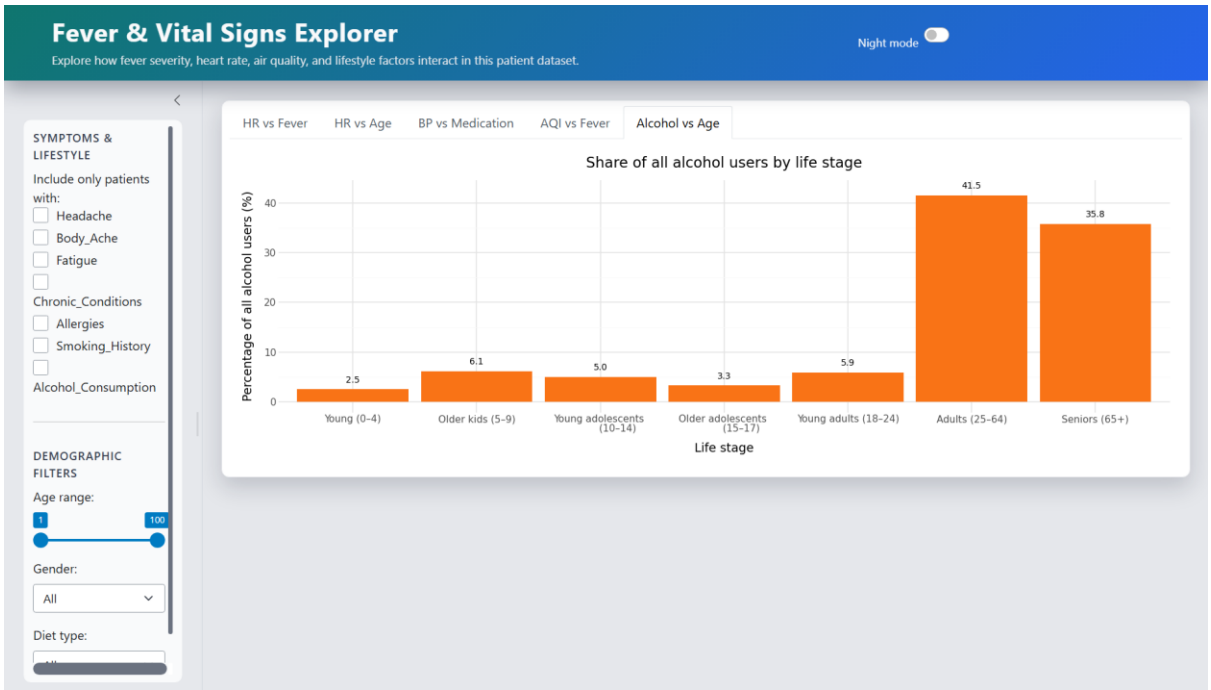
Column Name	Description	Data Type
Temperature	Body temperature of the patient in Celsius (e.g., 36.5 - 40.0).	Float
Fever Severity	Categorized fever severity: Normal, Mild Fever, High Fever.	Categorical
Age	Age of the patient (1 - 100 years).	Integer
Gender	Gender of the patient: Male or Female.	Categorical
BMI	Body Mass Index of the patient (e.g., 18.0 - 35.0).	Float
Headache	Whether the patient has a headache: Yes or No.	Categorical
Body Ache	Whether the patient has body aches: Yes or No.	Categorical
Fatigue	Whether the patient feels fatigued: Yes or No.	Categorical
Chronic Conditions	If the patient has any chronic conditions (e.g., diabetes, asthma): Yes or No.	Categorical
Allergies	If the patient has any allergies to medications: Yes or No.	Categorical
Smoking History	If the patient has a history of smoking: Yes or No.	Categorical
Alcohol Consumption	If the patient consumes alcohol: Yes or No.	Categorical
Humidity	Current humidity level in the patient's area (e.g., 30 - 90%).	Float
AQI	Current Air Quality Index in the patient's area (e.g., 0 - 500).	Integer
Physical Activity	Daily physical activity level: Sedentary, Moderate, Active.	Categorical
Diet Type	Diet preference: Vegetarian, Non-Vegetarian, or Vegan.	Categorical
Heartrate	Resting heart rate of the patient in beats per minute (e.g., 60 - 100).	Integer
Blood Pressure	Blood pressure category: Normal, High, or Low.	Categorical

Previous Medication	Medication previously taken by the patient: Paracetamol, Ibuprofen, Aspirin, or None.	Categorical
Recommended Medication	Target variable indicating the recommended medicine: Paracetamol or Ibuprofen.	Categorical

A2) Why the dataset is suitable for interactive visualization?

This dataset fits interactive visualization nicely because it has many different patient details like age, fever, BP, habits, AQI, etc. not just one or two numbers. When we add filters and sliders, we can quickly see how patterns change for different groups of patients instead of looking at one static chart.

B. Explanation of the App



The app layout has three key parts:

- **Top gradient header:** Shows the title *"Fever & Vital Signs Explorer"*, a short description, and a **Night mode** switch (dark/light theme).
- **Left sidebar:** Contains symptom and demographic filters: checkboxes for lifestyle/symptom flags and controls for age range, gender, and diet type.
- **Main content area:** A card with multiple tabs:
  - HR vs Fever
  - HR vs Age
  - BP vs Medication
  - AQI vs Fever
  - Alcohol vs Age

The typical workflow is:

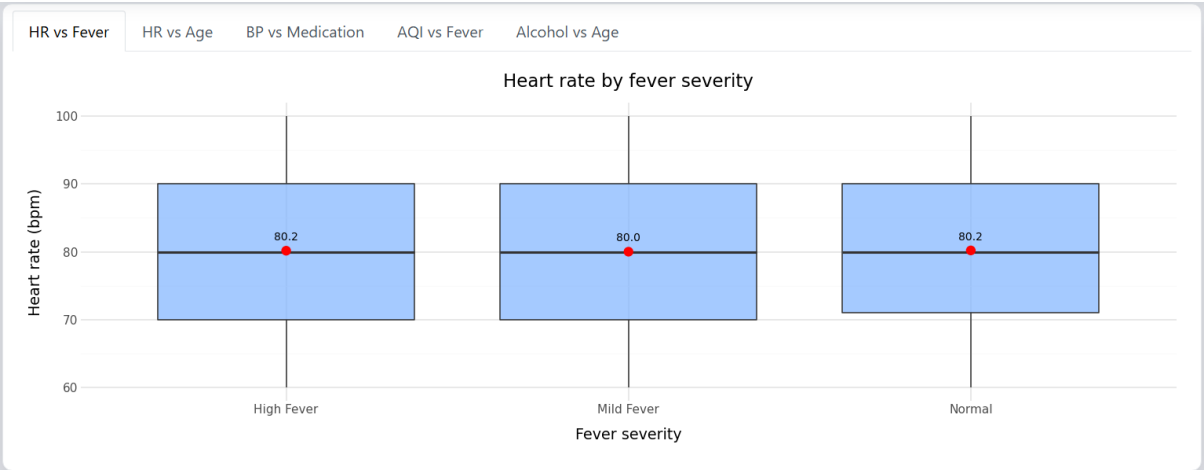
1. Select filters in the sidebar (age range, gender, diet, symptoms).
2. Navigate between tabs.
3. Observe how the plots update automatically as filters change.

5. Tasks and Screens (with explanations)

Objectives:

1. Show how heart rate changes with different fever levels.
2. Compare heart rate across simple age groups (young adults, adults, seniors).
3. See how blood pressure categories relate to the medicines recommended.
4. Check how AQI behaves for different fever severities.
5. Look at alcohol consumption patterns across life stages.
6. Let users filter by age, gender, diet, and symptoms to focus on specific groups.
7. Give a simple light/dark "Night mode" so the app is easy to read anywhere.

5.1 HR vs Fever:

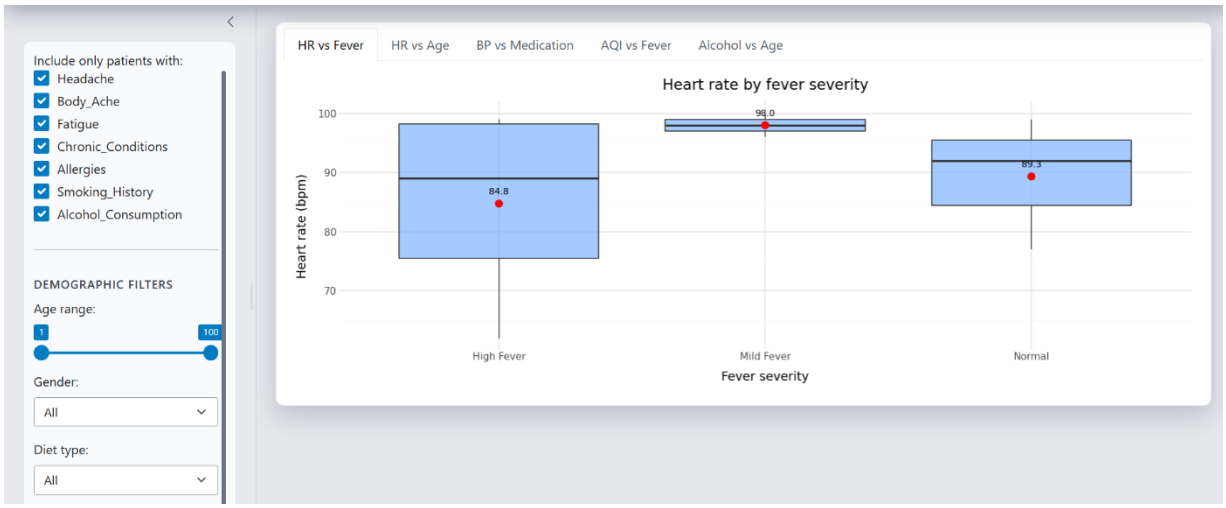


Action: Compare heart rate distributions
Target: Different fever severity levels

From this plot

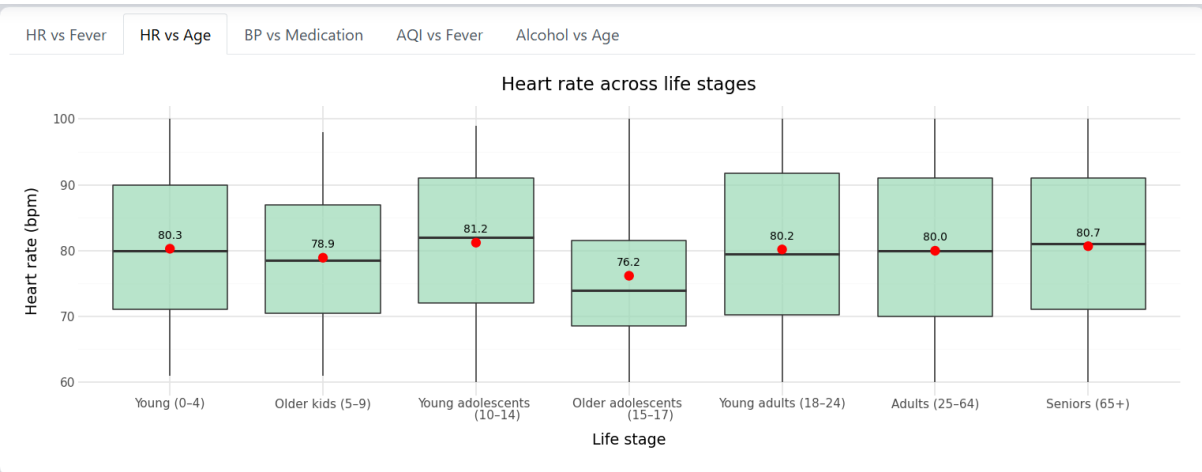
I. I can see how heart rate changes as fever severity increases.
II. I can compare the **spread** and **average** heart rate for each fever level.
III. I can check if higher fever is generally linked with higher heart rate in this dataset.

After Enabling Filter:



1) With all the filter options enabled u can see the changes that occur according to the data that we have.

5.2 HR vs Age (Life stages) plot

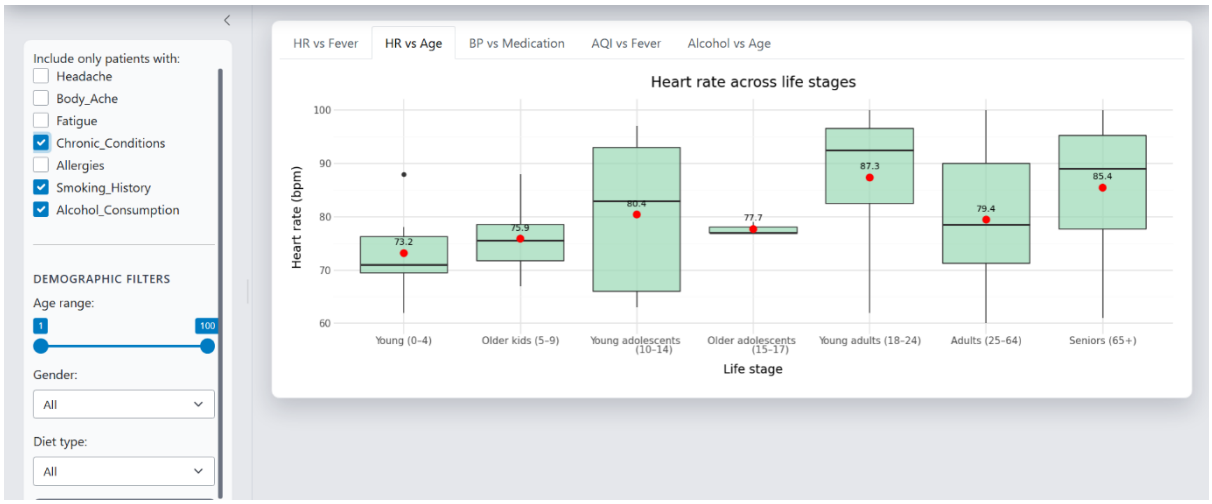


Action: Compare heart rate across age groups
Target: Life-stage categories (children, adolescents, young adults, adults, seniors)

From this plot

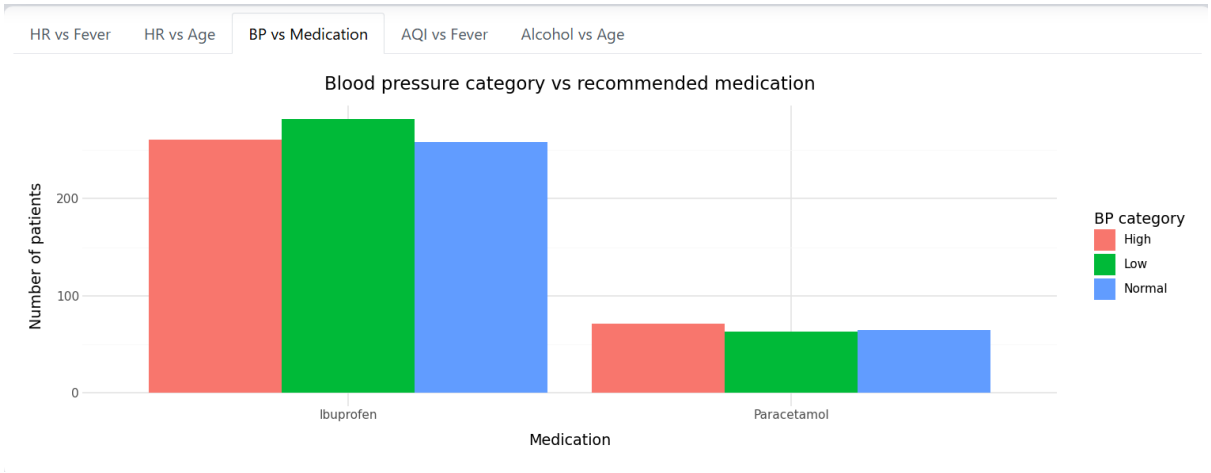
I. I can see which age groups tend to have higher or lower heart rates.
II. I can compare heart rate patterns between children, adolescents, young adults, adults, and seniors.
III. I can check if any life stage stands out as more “risky” in terms of heart rate.

After Enabling Filter:



1) we can see the changes In plot after enabling filters chronic_conditions, smoking_history, and alcohol_consumption to “yes”.

5.3BP vs Medication plot



Action: Compare blood pressure categories across treatments
Target: Recommended medications given to patients

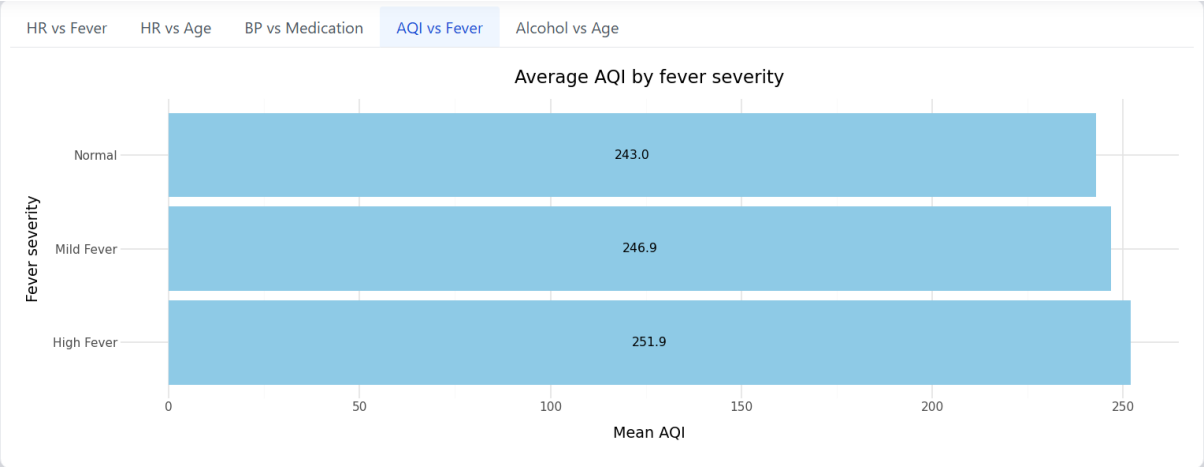
From this plot

I. I can see how different blood pressure categories are distributed across medications.
II. I can check which medications are mostly given to high BP vs normal BP patients.
III. I get a quick idea if treatment choices change with BP category.

After Enabling Filter

1) After toggling symptoms and lifestyle , we can see patients with normal bp, normal patients are recommended to take ibuprofen whereas, those with high bp are recommended to take Paracetamol.

5.4 AQI vs Fever plot

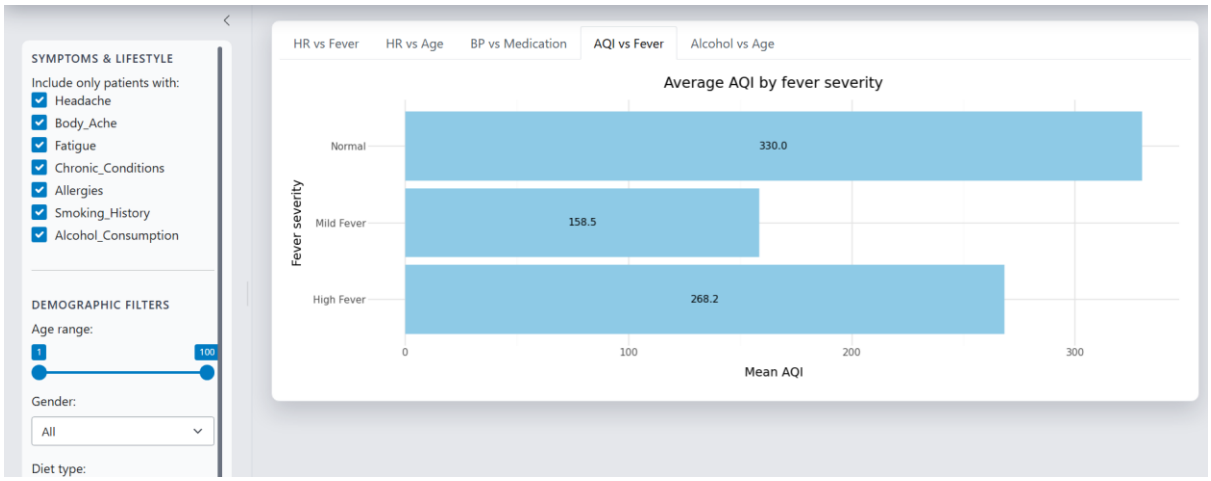


Action: Analyze average air quality
Target: Each fever severity level

From this plot

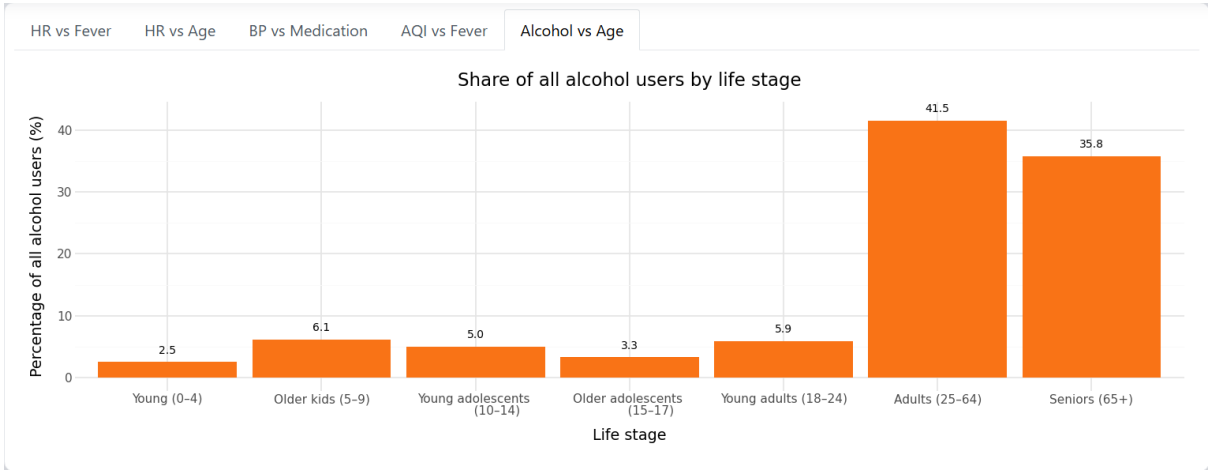
I. I can see how average air quality (AQI) varies across fever severity levels.
II. I can check if more severe fever cases are associated with higher or lower AQI in this dataset.
III. It gives a quick link between environmental condition (AQI) and fever severity.

After Enabling Filter



1) After turning on all the symptom and lifestyle filters, the app now shows **only patients who have every selected issue** (headache, fatigue, chronic conditions, smoking, alcohol, etc.).

5.4 Alcohol vs Age (Life stages) plot

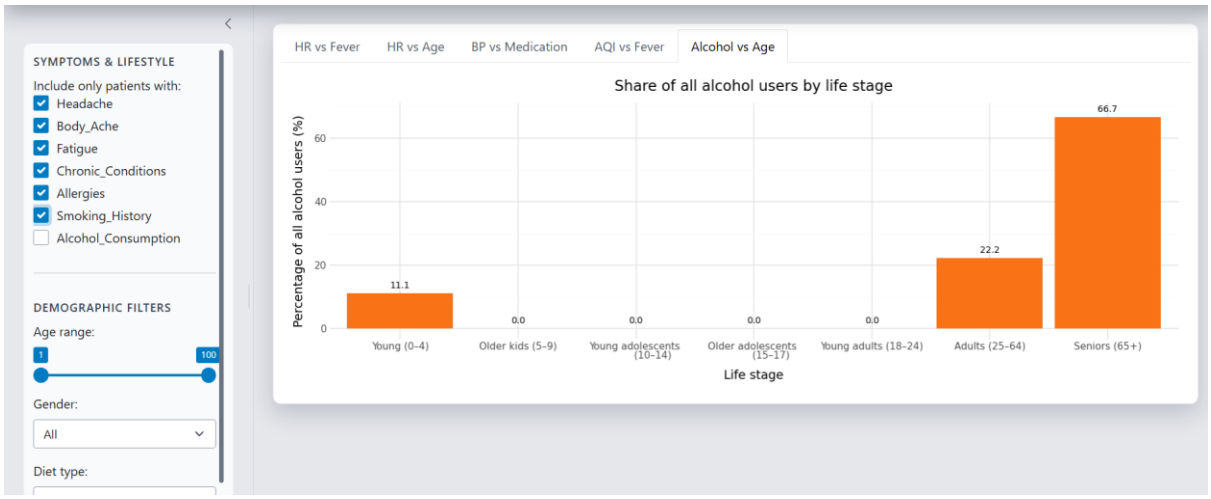


Action: Show distribution of alcohol users
Target: Different life-stage groups

From this plot:

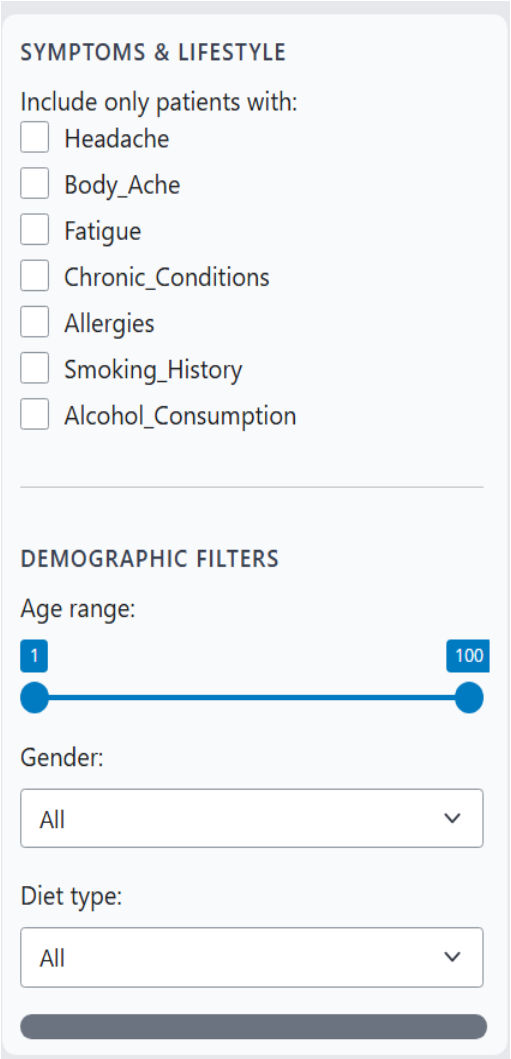
I. I can see which age groups contribute most to alcohol consumption.
II. I can check whether alcohol use is concentrated in young adults, adults, or spread across all stages.
III. This helps connect lifestyle behavior (alcohol) with demographic groups.

After Enabling Filter



Here, all symptom boxes and Smoking History are checked, so the app is showing only patients who are smokers with multiple symptoms.
In this filtered group, the Alcohol vs Age chart shows that almost all alcohol users are seniors (about 67%) and adults (about 22%), with only a small share in the youngest group and none in the adolescent ranges

5.5 Filtering



Task: Filter by **symptoms and lifestyle**: Tick boxes like headache, chronic condition, smoking, or alcohol to show only those patients and see the graphs update.

Action: Filter patients using symptom & lifestyle checkboxes.

Target: Patients who have the selected symptoms or habits.

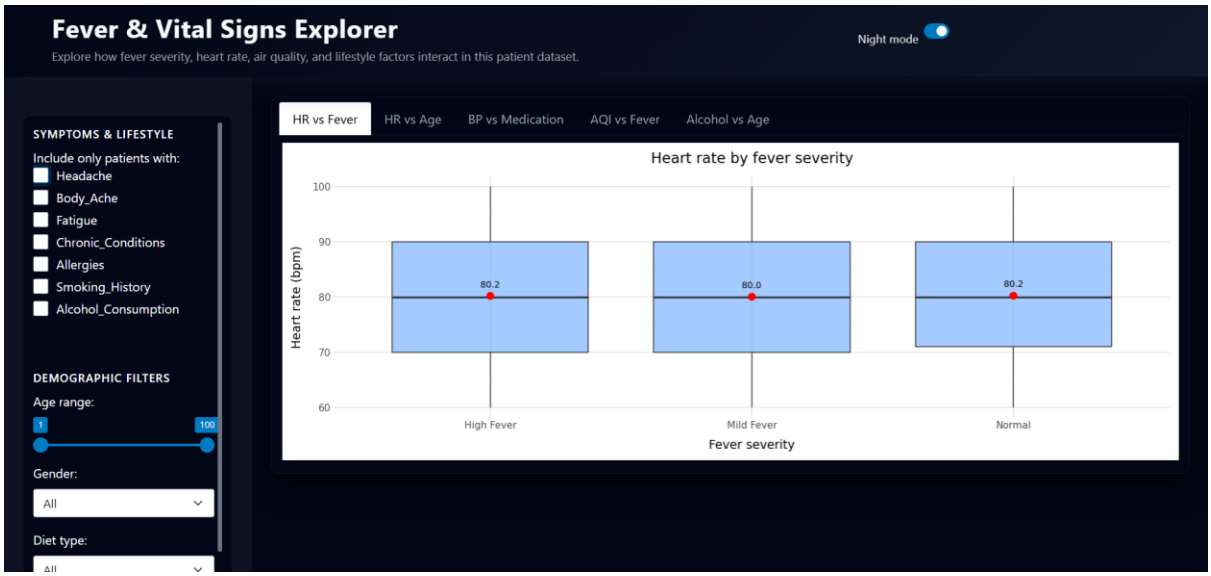
Task: Focus on **demographic groups**: Move the age slider and choose gender and diet to look at one group (for example, women aged 20 - 40),
all charts then show only that group.

Action: Filter patients using age slider, gender, and diet dropdowns.

Target: A specific demographic group
for example: women aged 20 - 40, vegetarians

5.6 Night mode:

The Night mode toggle switches the entire app from a light theme to a dark theme. It does not change the data or filters, but it makes the dashboard easier to read in low light or during presentations, while all plots and controls stay the same.



If you need action / target style:

- Task: Adjust the visual theme for better readability.
    - **Action:** Turn the Night mode switch on or off.
    - **Target:** Overall app appearance (background, text, and plots).